Theorem 18: hardness of approximating the optimal-transport cut distance

Correspondence between the paper and the Lean 4/Mathlib formalisation

Paper. J. Breitung, T. Gervens, M. Grohe. *Graph Similarity Meets Optimal Transport* (file `Graph_Similarity_Meets_Optimal_Transport.pdf` in the repository). Statement, equation, page and line numbers refer to this PDF; `docs/NUMBERING.md` lists the Lean files and declarations of all numbered statements.

The folder `GsimOT/Theorem22_CutHardness` formalises the reduction of Theorem 18 (p. 13, l. 443–444; proof in App. G, pp. 57–62) from MaxCut with Claims 58–61 and Eq. (47). It gives Theorem 4 (3) up to the complexity side. Shared notions come from `GsimOT/Common`; they are listed at the end.

Checking. Run `bash build.sh` in the repository root (see `README.md`). The script builds the library and runs `scripts/CheckAxioms.lean`, which must report only the standard axioms `propext`, `Classical.choice`, `Quot.sound`: no `sorry` and no additional axioms.

Main statement in Lean. `GsimOT.Hardness.maxcut_dOTCut_bounds_paper`: for every $0 < \delta < 1$ and every graph F on n vertices with an edge, the graphs G, H obtained from the base graphs G', H' by attaching $\ell_i = i(n^2+n)\lceil 1600n^6/\delta\rceil$ leaves to the i-th vertex satisfy

$$2(1-\delta)\,\mathrm{maxcut}(F) \le d_{\square}^{\mathrm{OT}}(G,H) \le 2\,\mathrm{maxcut}(F).$$

`GsimOT.Hardness.maxcut_dOTCut_bounds_eq47`: the case $\delta = 1/34$, Eq. (47): $\frac{33}{17}\,\mathrm{maxcut}(F) \le d_{\square}^{\mathrm{OT}}(G,H) \le 2\,\mathrm{maxcut}(F)$.

Correspondence

Paper	Page	Statement	Lean declaration(s) and location
Proof of Thm. 18 ([GG22], [AN06])	57, 59	Base graphs G', H' from the [AN06] matrix (G': edges of the positive entries, H': edges of the negative entries, l. 1885–1891) with $C_{\square}(\pi) = \lVert A_{G'} - A_{H'}\rVert_{\square} = 2\,\mathrm{maxcut}(F)$ (l. 1826–1830)	`GsimOT.Hardness.anG` `Theorem22_CutHardness/BaseGraphs.lean:92` `GsimOT.Hardness.anH` `Theorem22_CutHardness/BaseGraphs.lean:96` `GsimOT.Hardness.cutNorm_anG_sub_anH` `Theorem22_CutHardness/BaseGraphs.lean:218`
Proof of Thm. 18: leaves	57, 62	$\ell_i = i(n^2+n)k$ leaves at the i-th base vertex of both graphs (l. 1831–1834), $k = \lceil 1600n^6/\delta\rceil$ (l. 1977)	`GsimOT.Hardness.leafCount` `Theorem22_CutHardness/Theorem22.lean:107` `GsimOT.Hardness.leavesPaper` `Theorem22_CutHardness/Theorem22.lean:548`
Claim 58	58–59	If $Q(v, L_H) \ge \eta$ for a base vertex v, then $\lVert B(G,H,Q)\rVert_{\square} \ge \eta(M-2)/4$, where every base vertex has at least M leaves	`GsimOT.Hardness.claim56` `Theorem22_CutHardness/Claim56_LeafMass.lean:27`

Paper	Page	Statement	Lean declaration(s) and location
Claim 59	59–60	If $Q(v,w), Q(v,w') \geq \tau$ for distinct base vertices w, w', then $\|B(G,H,Q)\|_\square \geq \tau(M-2d)/2$, where d bounds the degrees in H'	`GsimOT.Hardness.claim57` Theorem22_CutHardness/Claim57_SplitMass.lean:28

Claim 60	60	If v_i sends mass to w_j, $i \neq j$: $\|B(G,H,Q)\|_\square \geq Q(v_i,w_j)(\|\ell_j - \ell_i\| - d)$, split by the order of ℓ_i, ℓ_j	`GsimOT.Hardness.claim58`
			Theorem22_CutHardness/Claim58_WrongVertex.lean:47
			`GsimOT.Hardness.claim58'`
			Theorem22_CutHardness/Claim58_WrongVertex.lean:78

Claim 61 (proof)	61	If $Q(v_i,w_i) \geq 1-\alpha$ for all i, then $\|B(G,H,Q)\|_\square \geq \|A_{G'} - A_{H'}\|_\square - 2\alpha\|V(G')\|^2$	`GsimOT.Hardness.claim59` Theorem22_CutHardness/Claim59_NearIdentity.lean:32
Claim 61	61	For $\delta \geq 0$, $0 \leq \alpha \leq \delta/(2n')^2$ and $Q(v_i,w_i) \geq 1-\alpha$ for all i: $\|B(G,H,Q)\|_\square \geq 2(1-\delta)\,c$ if $\|A_{G'} - A_{H'}\|_\square = 2c$, $c \in \mathbb{N}$	`GsimOT.Hardness.claim59_delta` Theorem22_CutHardness/Claim59_NearIdentity.lean:97

Facts used in Claims 58–60	59, 60, 62	$\Delta(G'), \Delta(H') \leq n-1$ (l. 1885–1892); $n' = 2m+n \leq n^2$ (l. 1969); $\mathrm{maxcut}(F) \leq n^2/4$ (l. 1880)	`GsimOT.Hardness.card_adj_anG_le`
			Theorem22_CutHardness/Theorem22.lean:426
			`GsimOT.Hardness.card_adj_anH_le`
			Theorem22_CutHardness/Theorem22.lean:456
			`GsimOT.Hardness.two_mul_card_edge_add_le`
			Theorem22_CutHardness/Theorem22.lean:487
			`GsimOT.Hardness.four_mul_maxcut_le`
			Theorem22_CutHardness/Theorem22.lean:511

Upper bound	58	$d_\square^{\mathrm{OT}}(G,H) \leq \|A_{G'} - A_{H'}\|_\square$ ($d_\square^{\mathrm{OT}} \leq d_\square$ and the canonical bijection, l. 1839–1840)	`GsimOT.Hardness.dOTCut_addLeaves_le` Theorem22_CutHardness/Theorem22.lean:111
Choice of the constants	61–62	For $0 < \delta < 1$: $2(1-\delta)c \leq d_\square^{\mathrm{OT}}(G,H)$ with $\alpha = \delta/(200n^4)$, $\eta = \alpha/2$, $\tau = \alpha/(4n^2)$, $k = \lceil 1600n^6/\delta \rceil$, for base graphs with $n' \leq n^2$, degrees $\leq n-1$ and $\|A_{G'} - A_{H'}\|_\square = 2c$, $1 \leq c \leq n^2/4$	`GsimOT.Hardness.le_dOTCut_addLeaves` Theorem22_CutHardness/Theorem22.lean:122
Theorem 18 (reduction)	13, 62	For $0 < \delta < 1$: $2(1-\delta)\,\mathrm{maxcut}(F) \leq d_\square^{\mathrm{OT}}(G,H) \leq 2\,\mathrm{maxcut}(F)$ for every graph F with an edge, with the construction of the paper (l. 1982–1983)	`GsimOT.Hardness.maxcut_dOTCut_bounds_paper` Theorem22_CutHardness/Theorem22.lean:555
Eq. (47)	58	$\frac{33}{17}\,\mathrm{maxcut}(F) \leq d_\square^{\mathrm{OT}}(G,H) \leq 2\,\mathrm{maxcut}(F)$: the case $\delta = 1/34$	`GsimOT.Hardness.maxcut_dOTCut_bounds_eq47` Theorem22_CutHardness/Theorem22.lean:593

Paper	Page	Statement	Lean declaration(s) and location
Theorem 18 (variant)	13	The reduction with n' in place of n in the leaf counts	GsimOT.Hardness.maxcut_dOTCut_bounds Theorem22_CutHardness/Theorem22.lean:320

Shared notions used (from GsimOT/Common)

Paper	Page	Statement	Lean declaration(s) and location
App. A	14	Cut norm $\|A\|_\square = \max_{S,T} \|A(S,T)\|$	GsimOT.cutNorm Common/CutNorm.lean:32 GsimOT.le_cutNorm Common/CutNorm.lean:39 GsimOT.cutNorm_le Common/CutNorm.lean:44 GsimOT.exists_eq_cutNorm Common/CutNorm.lean:52 GsimOT.cutNorm_smul Common/CutNorm.lean:59
Sec. 1, Sec. 2	2, 5	$B(G,H,Q)$ and $C_\square(Q) = \|B(G,H,Q)\|_\square$	GsimOT.B Common/OTDistances.lean:58 GsimOT.costCut Common/CutNorm.lean:79
Sec. 1, Sec. 2	1–2, 5	$d_\square$, $d_\square^{\mathrm{OT}}$ (infimum) and $\delta_\square = n^{-2} d_\square$, $\delta_\square^{\mathrm{OT}} = n^{-2} d_\square^{\mathrm{OT}}$	GsimOT.dCut Common/CutNorm.lean:82 GsimOT.dOTCut Common/CutNorm.lean:85 GsimOT.deltaCut Common/CutNorm.lean:88 GsimOT.deltaOTCut Common/CutNorm.lean:91

Paper	Page	Statement	Lean declaration(s) and location
App. E.2, Eq. (21), E.2.2	32, 36	$\zeta_R(A,B) = \|D_{A,B}\|_{\square,\mu_R}$ with $\mu_R = R/n$; $\zeta_R(A_G, A_H) = \zeta_R = n^{-2} C_\square(R)$ for graphs	`GsimOT.epsR` Common/CutNorm.lean:105 `GsimOT.adjMat` Common/CutNorm.lean:94 `GsimOT.epsR_adjMat` Common/CutNorm.lean:110
Sec. 1, App. B	1, 3, 15	$\kappa_{p\to q} = 1/q + 1 - 1/p$, $A_G^\pi - A_H$, $d_{p\to q}$ and $\delta_{p\to q} = n^{-\kappa_{p\to q}} d_{p\to q}$ $(s(n) = n^{\kappa_{p\to q}})$	`GsimOT.kappa` Common/AlignedDistances.lean:23 `GsimOT.alignedDiff` Common/AlignedDistances.lean:31 `GsimOT.distPQ` Common/AlignedDistances.lean:35 `GsimOT.deltaPQ` Common/AlignedDistances.lean:39
–	–	Sums over $V \times W$ against $Q \in \mathrm{DS}(V,W)$; partial sums $Q(S,T)$ and their bounds; $\mathrm{DS}(V,W)$ is compact; bounds for infima over $\mathrm{DS}(V,W)$	`GsimOT.IsDS.sum_mul_fst` Common/DoublyStochastic.lean:35 `GsimOT.mass` Common/DoublyStochastic.lean:118 `GsimOT.card_mul_le_of_lt_mass` Common/DoublyStochastic.lean:150 `GsimOT.isCompact_setOf_isDS` Common/DoublyStochastic.lean:77 `GsimOT.iInf_isDS_le` Common/DoublyStochastic.lean:98 `GsimOT.le_iInf_isDS` Common/DoublyStochastic.lean:104
Proofs of Thms. 15–17	50–57	Disjoint unions of paths (F_1) and of cycles; their degrees and edge counts	`GsimOT.Hardness.pathUnion` Common/PathsAndCycles.lean:56 `GsimOT.Hardness.cycleUnion` Common/PathsAndCycles.lean:226 `GsimOT.Hardness.sum_adj_pathUnion` Common/PathsAndCycles.lean:134 `GsimOT.Hardness.deg_cycleUnion` Common/PathsAndCycles.lean:238

Paper	Page	Statement	Lean declaration(s) and location
Proof of Thm. 15	50	Three-Partition instances $(A, a_1, \ldots, a_{3m})$; the step of adding 3 to each a_i and 9 to A	`GsimOT.Hardness.ThreePartition` Common/ThreePartition.lean:26 `GsimOT.Hardness.ThreePartition.IsYes` Common/ThreePartition.lean:39 `GsimOT.Hardness.ThreePartition.shift` Common/ThreePartition.lean:206 `GsimOT.Hardness.ThreePartition.shift_isYes_iff` Common/ThreePartition.lean:223
Proofs of Thms. 15–17	50–56	A solution gives a bijection mapping each path into one component; conversely a $Q \in \mathrm{DS}$ mapping every path into a single component gives a solution (counting mass)	`GsimOT.Hardness.ThreePartition.layoutEquiv` Common/ThreePartition.lean:137 `GsimOT.Hardness.ThreePartition.isYes_of_isDS` Common/ThreePartition.lean:168
Proofs of Thms. 17, 18	54, 57	Attaching $\ell(b)$ leaves to every vertex b of a base graph	`GsimOT.Hardness.addLeaves` Common/Leaves.lean:33

Notes on the formalisation

1. Only the correctness of the reduction is formalised: for the constructed graphs, YES-instances have a small and NO-instances a large distance. The complexity side (polynomial running time, NP-hardness of Three-Partition, Håstad's bound for MaxCut) is not formalised. The results of Section 4 and their shared gadgets are in the namespace `GsimOT.Hardness`.

2. The base graphs G', H' are built in `BaseGraphs.lean` from the [AN06] matrix as described in the proof of Claim 59 (p. 59, l. 1885–1891): the vertices are the signed edges $e^{\pm}$ and the vertices of F; for $e = uv$, $u < v$, G' has the edges e^+u, e^-v and H' the edges e^+v, e^-u. `cutNorm_anG_sub_anH` proves $\|A_{G'} - A_{H'}\|_\square = 2\,\mathrm{maxcut}(F)$, and the facts on G', H', F that the proof uses are proved: maximum degree at most $n-1$ (p. 59, l. 1885–1892), $n' = 2m+n \le n^2$ (p. 62, l. 1969) and $\mathrm{maxcut}(F) \le n^2/4$ (p. 59, l. 1880).

3. `maxcut_dOTCut_bounds_paper` is the reduction with the paper's leaf counts $i(n^2+n)\lceil 1600n^6/\delta\rceil$; `maxcut_dOTCut_bounds_eq47` is its case $\delta = 1/34$, Eq. (47). `maxcut_dOTCut_bounds` uses n' in place of n and needs neither the degree bound nor $n' \le n^2$.

4. Claims 58–61 are proved for general parameters (M, d, η, τ, ρ, α); the thresholds on k are checked once, in `le_dOTCut_addLeaves`, with the paper's $\alpha = \delta/(200n^4)$, $\eta = \alpha/2$, $\tau = \alpha/(4n^2)$ and $k = \lceil 1600n^6/\delta\rceil$. Claim 60 is split by the order of ℓ_i and ℓ_j. `claim59` bounds $\|B(G,H,Q)\|_\square$ from below by $\|A_{G'} - A_{H'}\|_\square - 2\alpha|V(G')|^2$; `claim59_delta` derives Claim 61 from it.